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Reaction 1

Similarly, elemental samples of alkaline-earth metals will react with elemental halogens (X_2) to form metal halides (Reaction 2), all of which are ionic except for $BeCl_2$.

$$\mathrm{M}(s) + \mathrm{X_2}(g) \longrightarrow \mathrm{MX_2}(s)$$

Reaction 2

Except for beryllium, the alkaline-earth metals react with water to produce very basic metal hydroxide solutions and hydrogen gas (Reaction 3).

$$\mathrm{M}(s) + 2\ \mathrm{H_2O}(l) \longrightarrow \mathrm{M(OH)_2}(aq) + \mathrm{H_2}(g)$$

Reaction 3

The most difficult alkaline-earth metal to study is radium, because it is a rare radioactive element that is produced from the nuclear decay of uranium. The most common natural isotope, radium-226, has a half-life of 1,600 years and decays by alpha emission into radon-222 (Reaction 4).

$$^{226}_{88}\text{Ra} \longrightarrow \, ^{222}_{86}\text{Rn} + \, ^{4}_{2}\text{He}$$

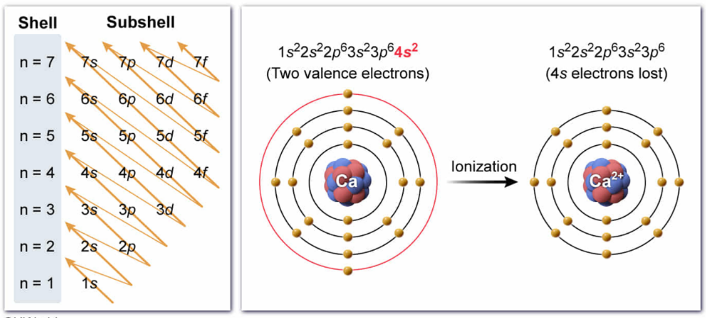
Reaction 4

Which of the following is the electron configuration of a Ca^{2+} ion?

- A. [Ne]$3s^2$ [1%]
- B. [Ne]$3s^2 3p^6$ [83%]
- C. [Ne]$3s^2 3p^6 4s^1$ [2%]
- D. [Ne]$3s^2 3p^6 4s^2$ [12%]

Correct
83% answered correctly

Explanation:



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An **electron configuration** accounts for the placement of all electrons within the **shells** and **subshells** of an atom. Shells are indicated using the principal quantum number (n), which relates to the distance of an orbital from the nucleus. Subshells are labeled by type (s, p, d, or f), which relate to the shape of the occupied orbital and hold a maximum of 2, 6, 10, and 14 electrons, respectively. Electron configurations are listed sequentially in order of **increasing energy**.

For elements with many electrons, a **noble gas abbreviation** may be used to express the electron configuration more concisely. This abbreviation uses the bracketed symbol for a preceding noble gas on the periodic table to denote the electron configuration for the corresponding number of lower-level electrons, followed by the configuration for the remaining higher-level electrons. For calcium, which has 20 electrons, the first 10 electrons have a configuration identical to that of neon ($1s^22s^22p^6$); therefore, [Ne] can be used as an abbreviation to denote this. This shortens the full electron configuration of calcium from $1s^22s^22p^63s^23p^64s^2$ to $[Ne]3s^23p^64s^2$. Alternatively, argon can be used instead to further shorten the configuration to $[Ar]4s^2$.

During ionization, the electrons in the highest energy shell/subshell are lost first. Calcium therefore loses its two valence electrons located in 4*s*. As a result, the electron configuration for the remaining electrons in the Ca^{+2} ion is $1s^22s^22p^63s^23p^6$, which can be abbreviated as $[Ne]3s^23p^6$.

(Choice A) $[Ne]3s^2$ is the electron configuration for a magnesium atom.

(Choice C) $[Ne]3s^23p^64s^1$ describes the electron configuration for a Ca^+ ion (which does not form).

(Choice D) $[Ne]3s^23p^64s^2$ gives the electron configuration for a calcium atom rather than for the Ca^{+2} ion.

Educational objective:

An electron configuration accounts for all electrons held by an atom or ion, listed sequentially by shell and subshell in order of increasing energy. A noble gas abbreviation may be used to represent the configuration of lower-level electrons. Ionized atoms lose electrons from higher energy levels first.

Time spent:0 sec
Subject:
General Chemistry

QID:401676
System:
Atomic Theory and Chemical Composition

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$$2\ \mathrm{M}(s) + \mathrm{O_2}(g) \longrightarrow 2\ \mathrm{MO}(s)$$

Reaction 1

Similarly, elemental samples of alkaline-earth metals will react with elemental halogens (X_2) to form metal halides (Reaction 2), all of which are ionic except for $BeCl_2$.

$$\mathrm{M}(s) + \mathrm{X_2}(g) \longrightarrow \mathrm{MX_2}(s)$$

Reaction 2

Except for beryllium, the alkaline-earth metals react with water to produce very basic metal hydroxide solutions and hydrogen gas (Reaction 3).

$$\mathrm{M}(s) + 2\ \mathrm{H_2O}(l) \longrightarrow \mathrm{M(OH)_2}(\mathrm{aq}) + \mathrm{H_2}(g)$$

Reaction 3

The most difficult alkaline-earth metal to study is radium, because it is a rare radioactive element that is produced from the nuclear decay of uranium. The most common natural isotope, radium-226, has a half-life of 1,600 years and decays by alpha emission into radon-222 (Reaction 4).

$${}^{226}_{88}\mathrm{Ra} \longrightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

Reaction 4

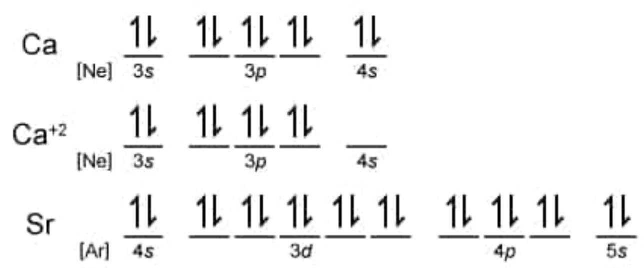
Which of the following alkaline-earth species are predicted to be diamagnetic based on their electron configuration? (Note: Assume that the species are single, isolated atoms that are not part of a bulk metal sample.)

I. Ca
II. Ca^{+2}
III. Sr

- A. I only [1%]
- B. II only [17%]
- C. I and III only [23%]
- ✔ D. I, II, and III [58%]

Correct
57% answered correctly

Explanation:



Ca, Ca^{+2}, and Sr atoms are predicted to be diamagnetic because their electron configurations have no unpaired electrons.

Electrons fill the shells and subshells of an atom in order of increasing energy as described by the atom's electron configuration. As stated by the Pauli exclusion principle, each orbital within a subshell can hold a maximum of two electrons, which must have opposite spins. In schematic representations, these electrons are represented by arrows and the spin is indicated by the orientation of the arrow (up or down). The occupied orbitals are often represented by blanks or boxes. Hund's rule states that electrons fill orbitals in such a way as to maximize the number of unpaired electrons.

As a sublevel is filled, electrons form pairs only after all orbitals contain at least one electron. The magnetic properties of an atom (or ion) depend on whether or not unpaired electrons remain after all electrons are assigned to an orbital. If **unpaired electrons** remain, then the atom is **paramagnetic** and the unpaired electrons will be attracted to an external magnetic field. If all electrons are paired, then the atom is **diamagnetic** and the paired electrons will be repelled by a magnetic field.

The electron configurations of the alkaline-earth metals end with a completely filled *s*-block. Therefore, the electrons occupying the corresponding *s* orbital in the respective valence shells are paired. Accordingly, isolated atoms of Ca and Sr have no unpaired electrons and are both predicted to be diamagnetic **(Numbers I and III)**. The Ca^{+2} ion results from losing both electrons from the valence shell, but because all the shells below the valence shell are fully filled

(no unpaired electrons), an isolated Ca^{+2} ion is also predicted to be diamagnetic **(Number II)**.

Interestingly, experimental measurements of bulk metal samples of calcium and strontium disagree with these predictions because these predictions are for single, isolated atoms. In metallic solids made of many moles of atoms, the valence electrons are delocalized throughout the sample and become unpaired, which results in paramagnetic properties.

Educational objective:
Electrons fill the shells and subshells of an atom in order of increasing energy according to Hund's rule and the Pauli exclusion principle. Unpaired electrons in the electronic configuration of an atom or ion result in paramagnetism, but a configuration without unpaired electrons results in diamagnetism.

Time spent:0 sec
Subject:
General Chemistry

QID:401677
System:
Atomic Theory and Chemical Composition

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$$2\,\mathrm{M}(s) + \mathrm{O_2}(g) \longrightarrow 2\,\mathrm{MO}(s)$$

Reaction 1

Similarly, elemental samples of alkaline-earth metals will react with elemental halogens (X_2) to form metal halides (Reaction 2), all of which are ionic except for $BeCl_2$.

$$\mathrm{M}(s) + \mathrm{X_2}(g) \longrightarrow \mathrm{MX_2}(s)$$

Reaction 2

Except for beryllium, the alkaline-earth metals react with water to produce very basic metal hydroxide solutions and hydrogen gas (Reaction 3).

$$\mathrm{M}(s) + 2\,\mathrm{H_2O}(l) \longrightarrow \mathrm{M(OH)_2}(\mathrm{aq}) + \mathrm{H_2}(g)$$

Reaction 3

The most difficult alkaline-earth metal to study is radium, because it is a rare radioactive element that is produced from the nuclear decay of uranium. The most common natural isotope, radium-226, has a half-life of 1,600 years and decays by alpha emission into radon-222 (Reaction 4).

$$^{226}_{88}\mathrm{Ra} \longrightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

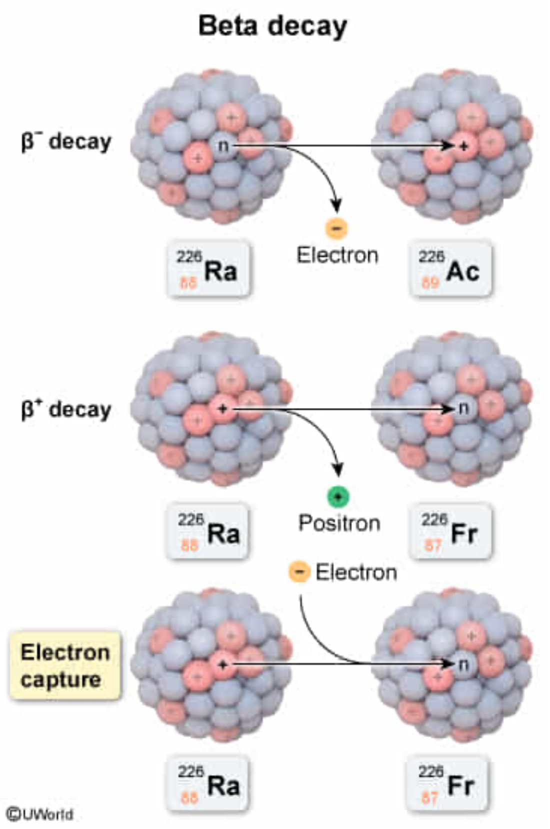
Reaction 4

If radium-226 were to undergo radioactive decay by electron capture (a type of beta decay) instead of by alpha emission, the resulting nucleus would be:

- A. $^{222}_{86}\mathrm{Rn}$ [5%]
- ✔ B. $^{226}_{87}\mathrm{Fr}$ [58%]
- C. $^{226}_{88}\mathrm{Ra}$ [10%]
- D. $^{226}_{89}\mathrm{Ac}$ [25%]

Correct
57% answered correctly

Explanation:



Radioactive **beta decay** can occur in three forms: β^--decay (electron emission), β^+-decay (**positron emission**), and **electron capture**. In **β^--decay**, a neutron converts into a nuclear proton and emits an electron. In **β^+-decay**, a proton converts into a neutron (the opposite of β^--decay) and emits a positron (ie, an electron with a positive charge). In electron capture, a proton captures an electron near the nucleus and converts into a neutron without a positron or electron emission.

In all three forms of beta decay, the mass number remains unchanged, while the atomic number increases (β^--decay) or decreases (β^+-decay and electron capture) by 1. As the atomic number changes, the identity of the element changes accordingly.

If radium-226 underwent an electron capture, the result would yield a nucleus of the same mass number (226) but an atomic number that is 1 less ($88 - 1 = 87$). Therefore, francium-226 would be detected.

(Choice A) Radon-222 has a mass number that is decreased by 4 and an atomic number that is decreased by 2 relative to the radium-226 isotope. This is the result of alpha decay rather than β^+-decay.

(Choice C) If the mass number and atomic numbers are unchanged, beta decay has not occurred. This outcome would indicate either no nuclear conversion or the release of nuclear energy as a gamma emission.

(Choice D) Actinium-226 would be the result of β^--decay (electron emission), not the result of an electron capture.

Educational objective:
In all forms of beta decay, the mass number remains unchanged, while the atomic number increases (β^--decay) or decreases (β^+-decay and electron capture) by 1. β^--decay converts a neutron into a proton and emits an electron. β^+-decay and electron capture convert a proton into a neutron (the opposite of a β^--decay).

Time spent: 0 sec	**QID**:401678
Subject: General Chemistry	**System:** Atomic Theory and Chemical Composition

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$$\mathrm{M}(s) + \mathrm{X_2}(g) \longrightarrow \mathrm{MX_2}(s)$$

Reaction 2

Except for beryllium, the alkaline-earth metals react with water to produce very basic metal hydroxide solutions and hydrogen gas (Reaction 3).

$$\mathrm{M}(s) + 2\,\mathrm{H_2O}(l) \longrightarrow \mathrm{M(OH)_2}(aq) + \mathrm{H_2}(g)$$

Reaction 3

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$${}^{226}_{88}\mathrm{Ra} \longrightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

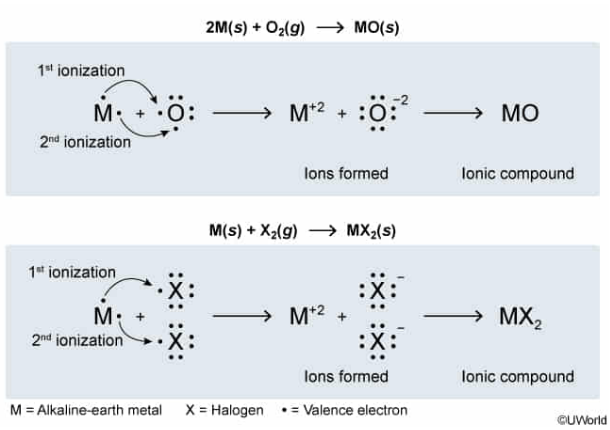
Reaction 4

When forming ionic bonds with nonmetals (Reactions 1-3), the reactivity of the alkaline-earth metals is observed to increase moving down the column. Based on atomic properties, this trend in reactivity is best explained by comparing:

- ✔ A. the energy required to remove an electron from each metal atom. [61%]
- B. the tendency of each metal atom to attract electrons within a bond. [11%]
- C. the extent to which the electron cloud of each metal atom can be distorted by an external charge. [16%]
- D. the energy released when an electron is added to each metal atom. [9%]

Correct
62% answered correctly

Explanation:



Many atomic properties are associated with electrons. Four such properties are polarizability, electronegativity, electron affinity, and ionization energy.

- Polarizability is the extent to which an electron cloud of an atom can be distorted by an external charge or by an applied electric field to produce a dipole.
- Electronegativity is the tendency of an atom to attract electrons within a bond.
- Electron affinity assesses the tendency of an atom to accept an additional electron by measuring the energy change when an electron is added to an atom.
- Ionization energy (the opposite of electron affinity) measures the **energy** required to **remove an electron** from an atom.

Reactions 1-3 involve the formation of an ionic bond between an alkaline-earth metal and a nonmetal. This requires the valence electrons of the metal atom to be removed and transferred to the nonmetal atom. Of the four properties previously discussed, only ionization energy quantifies how easily electrons are removed from an atom. Therefore, ionization energy best explains the reactivity trend for Reactions

1-3. Moving down the alkaline-earth metal column, ionization energy decreases. This makes removing an electron more favorable and increases reactivity.

(Choice B) Although the reactivity of the alkaline-earth metals does increase as electronegativity decreases, this correlation does not explain the *cause* of the reactivity trend. Reactions 1-3 proceed by forming ions (ionization); therefore, electronegativity is not the best property to explain the reactivity trend.

(Choice C) Reactions 1-3 depend on the transfer of electrons (ionic bond formation) rather than on the distortion of the electron cloud.

(Choice D) Electrons must be removed from (not added to) the alkaline-earth metals for Reactions 1-3 to occur.

Educational objective:
Ionization energy is the energy required to remove an electron from an atom. Because of the associated ionization and electron transfer involved in forming ionic bonds, the reactivity of atoms forming ionic compounds increases as the ionization energy decreases.

Time spent:0 sec
Subject:
General Chemistry

QID:401679
System:
Atomic Theory and Chemical Composition

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$$\mathrm{M}(s) + \mathrm{X_2}(g) \longrightarrow \mathrm{MX_2}(s)$$

Reaction 2

Except for beryllium, the alkaline-earth metals react with water to produce very basic metal hydroxide solutions and hydrogen gas (Reaction 3).

$$\mathrm{M}(s) + 2\,\mathrm{H_2O}(l) \longrightarrow \mathrm{M(OH)_2}(aq) + \mathrm{H_2}(g)$$

Reaction 3

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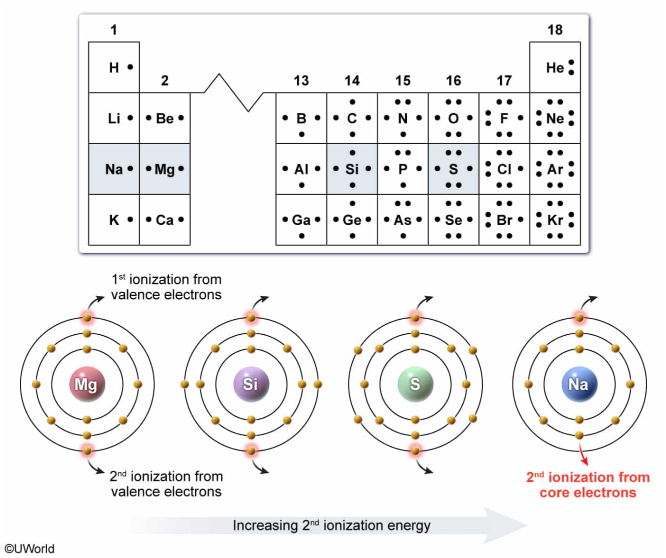
Reaction 4

When comparing the second ionization energies of the atoms across the same row of the periodic table, which of the following groups will have the atom with the lowest value?

- A. Group 1 [37%]
- B. Group 2 [46%]
- C. Group 14 [4%]
- D. Group 16 [11%]

Correct
46% answered correctly

Explanation:



Ionization energy is the energy required to remove an electron (e^-) from an atom or ion (measured in the gas state). The first ionization energy refers to the removal of the first, most loosely bound electron from a neutral atom of an element (X):

$$X + \text{energy} \rightarrow X^+ + e^-$$

The **second ionization energy** refers to the **removal** of a **second electron**:

$$X^+ + \text{energy} \rightarrow X^{+2} + e^-$$

Valence electrons are more loosely bound than core electrons and are removed first. Accordingly, **removing** a **core electron** takes **more energy**

than does removing a **valence electron**. On the periodic table, metals generally have a lower first ionization energy than nonmetals, and first ionization energies tend to increase going left to right across a row and decrease going down a column. The same is true for the second ionization energy except when the second electron being removed is not a *valence electron*.

If comparing elements in the third period (row), the nonmetals Si (Group 14) and S (Group 16) are farther to the right in the period, and both have a higher second ionization energy than Mg. However, the second ionization energy of Na (Group 1) is much higher than that of Mg (Group 2) because Na has only one valence electron and removing a second electron from Na requires the loss of a *core electron*. The availability of a second valence electron causes the second ionization energy of Mg (Group 2) to have the lowest value.

(Choice A) Group 1 metals have only one valence electron, and a second ionization requires the loss of a tightly bound core electron.

(Choices C and D) When removing valence electrons, nonmetals generally have a higher second ionization energy than metals; among metals, ionization energies tend to increase from left to right across a period.

Educational objective:
The second ionization energy is the energy required to remove the second of two electrons from an atom. The second ionization energy tends to increase across a period and to decrease down a group; however, ionizations involving core electrons are higher energy than those involving valence electrons.

Time spent:0 sec
Subject:
General Chemistry

QID:401680
System:
Atomic Theory and Chemical Composition

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$$2\,\mathrm{M}(s) + \mathrm{O_2}(g) \longrightarrow 2\,\mathrm{MO}(s)$$

Reaction 1

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$$\mathrm{M}(s) + \mathrm{X_2}(g) \longrightarrow \mathrm{MX_2}(s)$$

Reaction 2

Except for beryllium, the alkaline-earth metals react with water to produce very basic metal hydroxide solutions and hydrogen gas (Reaction 3).

$$\mathrm{M}(s) + 2\,\mathrm{H_2O}(l) \longrightarrow \mathrm{M(OH)_2}(aq) + \mathrm{H_2}(g)$$

Reaction 3

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$$^{226}_{88}\mathrm{Ra} \longrightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

Reaction 4

What is the mass of water necessary to generate 11.2 L of hydrogen gas if calcium metal reacts with water at standard temperature and pressure (STP)?

A. 2.0 g [7%]

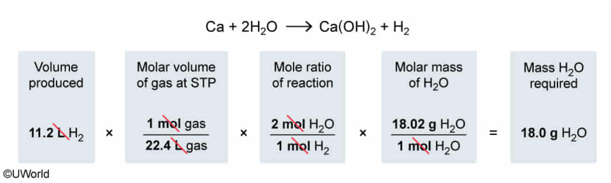
B. 4.5 g [16%]

C. 9.0 g [34%]

D. 18.0 g [42%]

Correct
42% answered correctly

Explanation:



Under conditions of **standard temperature and pressure (STP)**, which are defined as 0 °C (273 K) and 1.00 atm pressure, 1.00 mole of gas occupies a volume of 22.4 L. This molar volume (which is only valid at STP) is often just used as a known constant, but this value can be derived using the ideal gas law as follows:

$$PV = nRT$$

$$\frac{V}{n} = \frac{RT}{P} = \frac{\left(0.0821\ \frac{L \cdot \cancel{atm}}{mol \cdot \cancel{K}}\right)(273\ \cancel{K})}{(1.00\ \cancel{atm})} = 22.4\ L/mol$$

Using this value as a conversion factor, the moles of a gas present in a given volume at STP can be determined. If 11.2 L of H_2 gas are produced in the reaction of calcium metal with water (Reaction 3) at STP, the number of moles of H_2 gas produced in the reaction is given by:

$$11.2\ \cancel{L}\ H_2 \times \frac{1\ mol}{22.4\ \cancel{L}} = 0.500\ mol\ H_2$$

In a reaction, the moles of one species can be related to the moles of other chemical species by applying stoichiometric mole ratios obtained from a balanced reaction equation. According to the balanced equation of Reaction 3, the molar ratio of H_2O to H_2 is 2:1. Using this ratio, the required number of moles of H_2O can be determined:

$$0.500\ \cancel{mol\ H_2} \times \frac{2\ mol\ H_2O}{1\cancel{mol\ H_2}} = 1.00\ mol\ H_2O$$

The mass of the required moles of H_2O can be calculated using its **molar mass**:

$$1.00\ \cancel{\text{mol H}_2\text{O}} \times \frac{18.02\ \text{g H}_2\text{O}}{1\ \cancel{\text{mol H}_2\text{O}}} = 18.02\ \text{g H}_2\text{O}$$

(Choice A) 2.0 g is obtained if the calculation is done using the molar mass of H_2 (2.02 g/mol) instead of H_2O (18.02 g/mol).

(Choice B) 4.5 g is obtained if the 2:1 mole ratio of $H_2O:H_2$ is flipped and the 0.500 moles of H_2 is divided by 2 instead of multiplied by 2 during the second step of the calculation.

(Choice C) 9.0 g is obtained if a 1:1 mole ratio of $H_2O:H_2$ is used instead of the correct 2:1 ratio in the second step of the calculation.

Educational objective:
At standard temperature and pressure (STP), which are 0 °C (273 K) and 1.00 atm, gases occupy a molar volume of 22.4 L/mol. Mole ratios from balanced chemical reactions can be used to relate the moles of the gas to the moles of other reaction species, and the mass of a given number of moles of a particular species can be found using its molar mass.

Time spent:0 sec
Subject:
General Chemistry

QID:401681
System:
Atomic Theory and Chemical Composition